

Madison Shaw

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Skills

- Python, Java, C++, SQL, x86 Assembly, Racket, CircuitPython, PyQt
- Raspberry Pi, Microcontrollers, Soldering, Circuit Design
- Excel, Git, Onshape CAD, JSON, Ultralytics YOLO, CoralNet, VIAME

Experience

University of Rhode Island Graduate School of Oceanography | RI, Narragansett SURFO Research Fellow (NSF-REU) | 06/2025 - Present

- Trained and deployed two AI models: one for segmenting crystals in SEM images of volcanic rock samples and another for detecting various biology in underwater ROV footage.
- Utilized CoralNet-Toolbox and VIAME to annotate datasets and train models/pipelines
- Designed a GUI and analysis tool using PyQt to graph area and perimeter distributions, and generate major/minor axis scatterplots
- Assisted in lab work preparing volcanic rock samples for high-resolution SEM imaging

Medtronic | CT, North Haven Software Engineering Intern | 06/2024 - 07/2024

- Developed GUI with PyQt, utilizing signals and slots to dynamically update stapling device settings
- Designed JSON schema for data validation
- Applied the Factory Method design pattern to instantiate objects based on schema-defined formats
- Participated in daily stand-up meetings with software development team
- Earned Lean Six Sigma Yellow Belt certification

Medtronic | CT, North Haven Software Engineering Intern | 06/2023 - 08/2023

- Designed and built a custom testing device using a Raspberry Pi Pico, micro USB, and accelerometer
- Soldered hardware components and programmed microcontrollers using CircuitPython for data collection
- Developed and implemented experimental test methods for hardware validation
- Analyzed sensor data in Excel to determine accelerometer threshold values
- Led STEM outreach activities with elementary school students to promote hands-on learning

Education

Worcester Polytechnic Institute | Worcester, MA Computer Science and Business Analytics | 05/2027

- Dean's List (2023-2026)
- Involvements: Computer Science Peer Learning Assistant, Varsity Women's Soccer, Student Music Association (Treasurer), Alpha Xi Delta Women's Fraternity (Finance Director)

Certifications

- Lean Six Sigma Yellow Belt Certification
- SOLIDWORKS Design Associate (CSWA)